

ROLLING THE DICE

*Risk, Reward &
Decision-Making
in Software*



PAUL TURCHINETZ - QA OR THE HIGHWAY 2026



THE FUGU FISH

ROLLING THE DICE



WHAT ARE THE ODDS?

ROOMMATES

ROOMMATES ARE A GAMBLE...

ROOMMATES

- We roll a dice for each roommate

ROOMMATES ARE A GAMBLE...

ROOMMATES

- We roll a dice for each roommate
- Each face represents a day

ROOMMATES ARE A GAMBLE...

What are you coming home to?

What are you coming home to?

All Faces on the Weekly Dice:

Dishes

Clutter

Nothing

Nothing

Nothing

Free Food

Guests

What are you coming home to?

All Faces on the Weekly Dice:

Dishes

Clutter

Nothing

Nothing

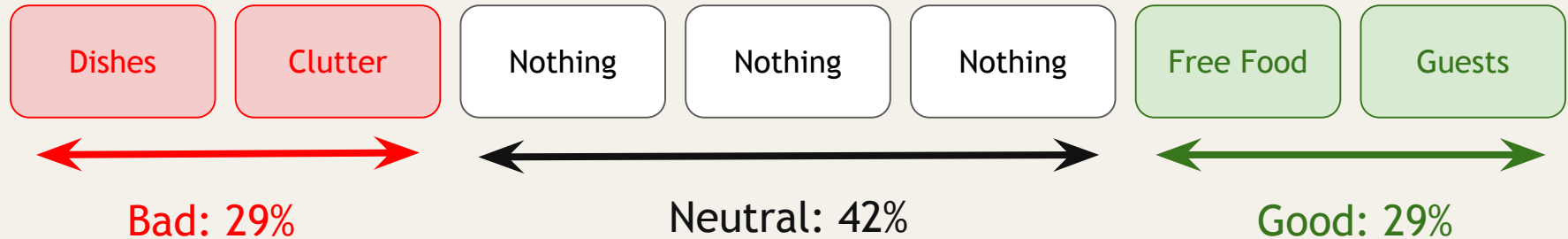
Nothing

Free Food

Guests

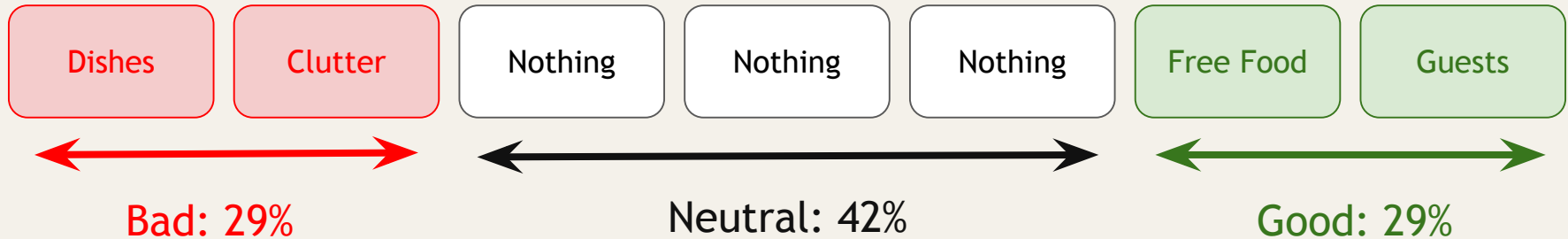
What are you coming home to?

All Faces on the Weekly Dice:



What are you coming home to?

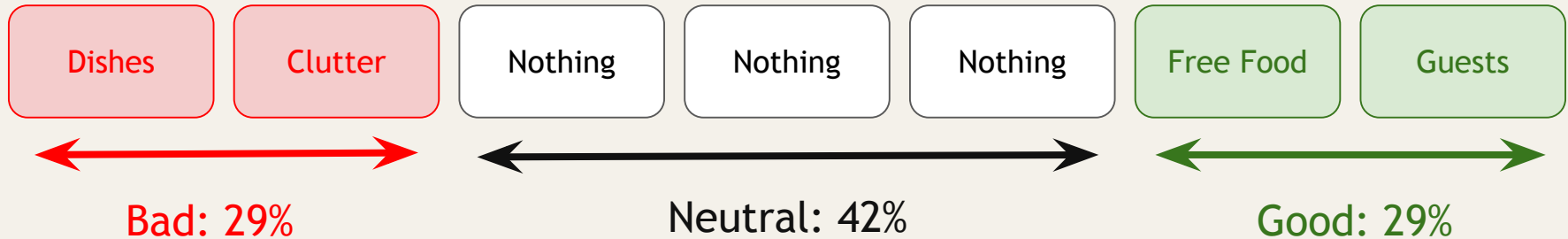
All Faces on the Weekly Dice:



Range?

What are you coming home to?

All Faces on the Weekly Dice:

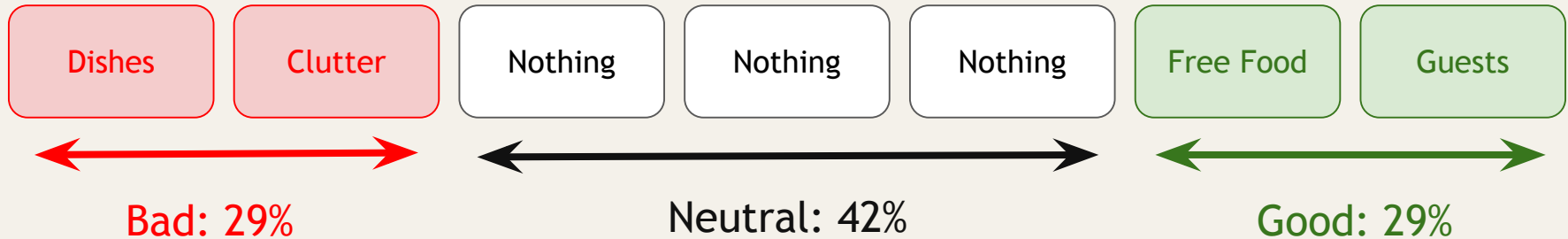


Range?

Expected Value?

What are you coming home to?

All Faces on the Weekly Dice:



Range?

Expected Value?

Consistency?

New Rule: No Parties

New Rule: No Parties

All Faces on the Weekly Dice:

Dishes

Nothing
(Clutter)

Nothing

Nothing

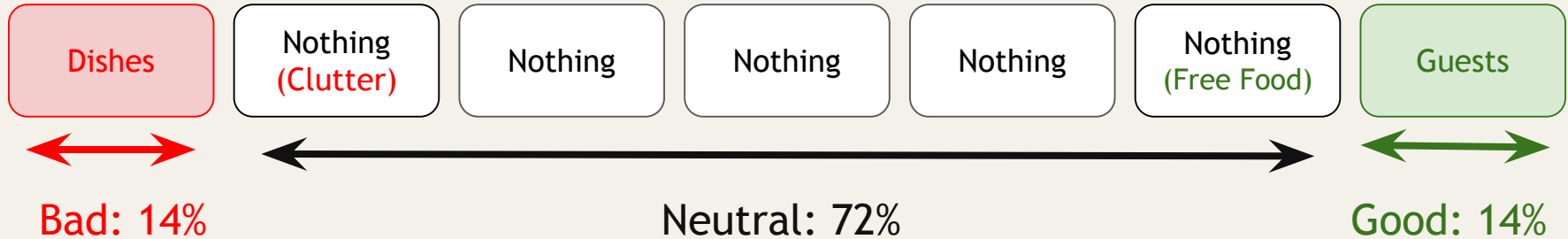
Nothing

Nothing
(Free Food)

Guests

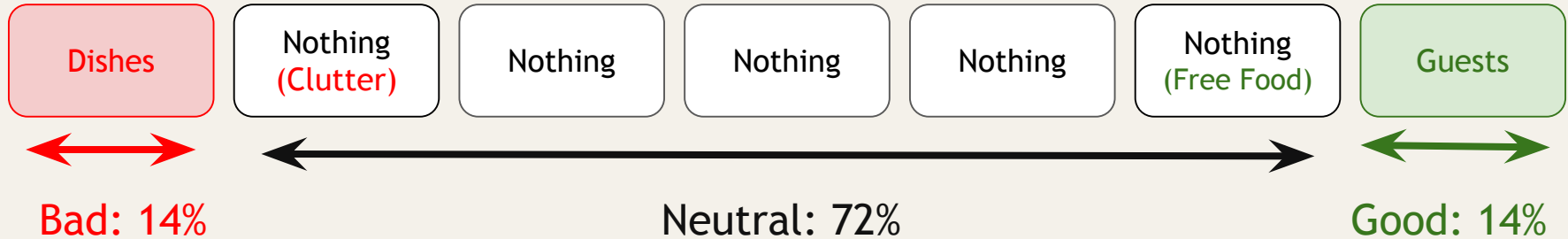
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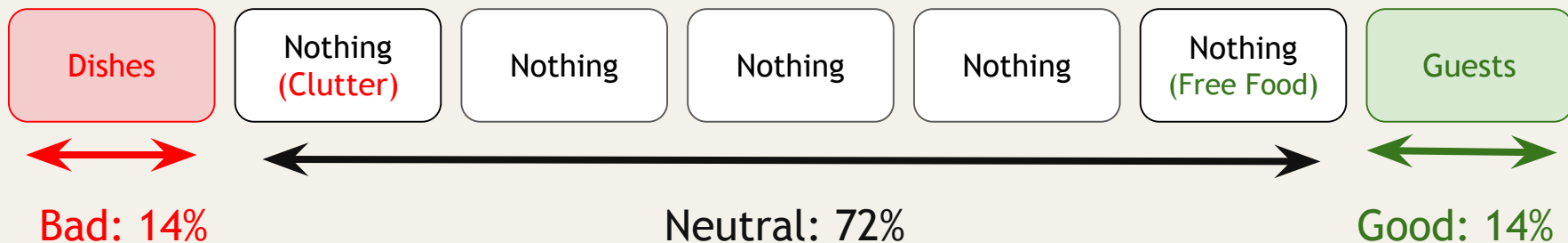
All Faces on the Weekly Dice:



Range?

New Rule: No Parties

All Faces on the Weekly Dice:

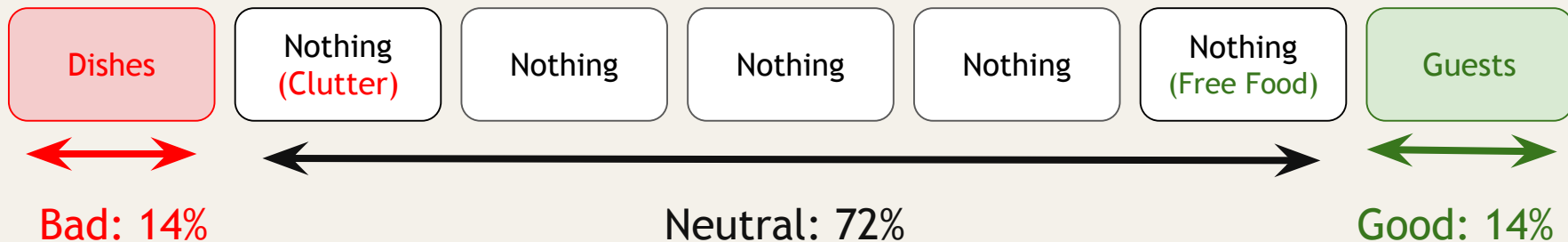


Range?

Expected Value?

New Rule: No Parties

All Faces on the Weekly Dice:



Range?

Expected Value?

Consistency?

Having Multiple Roommates:

Having Multiple Roommates:

A:

Dishes

Nothing
(Clutter)

Nothing

Nothing

Nothing

Nothing
(Free Food)

Guests

Having Multiple Roommates:

A:	Dishes	Nothing (Clutter)	Nothing	Nothing	Nothing	Nothing (Free Food)	Guests
B:	Nothing	Nothing (Free Food)	Dishes	Nothing	Nothing	Guests	Nothing (Clutter)
C:	Nothing	Nothing	Guests	Nothing	Nothing (Free Food)	Nothing (Clutter)	Dishes
D:	Nothing (Free Food)	Nothing (Clutter)	Dishes	Nothing	Guests	Nothing	Nothing

Having Multiple Roommates:

A:	Dishes	Nothing (Clutter)	Nothing	Nothing	Nothing	Nothing (Free Food)	Guests
B:	Nothing	Nothing (Free Food)	Dishes	Nothing	Nothing	Guests	Nothing (Clutter)
C:	Nothing	Nothing	Guests	Nothing	Nothing (Free Food)	Nothing (Clutter)	Dishes
D:	Nothing (Free Food)	Nothing (Clutter)	Dishes	Nothing	Guests	Nothing	Nothing

(Tuesday)



ROOMMATES

What are the Statistics of these Dice?

ROOMMATES

What are the Statistics of these Dice?

- How many *total* faces?



ROOMMATES

What are the Statistics of these Dice?



- How many *total* faces?
- How many *different* faces?

ROOMMATES

What are the Statistics of these Dice?



- How many *total* faces?
- How many *different* faces?
- Can you *calculate* Range, EV, Consistency?



ROOMMATES

What are the Statistics of these Dice?

- How many *total* faces?
- How many *different* faces?
- Can you *calculate* Range, EV, Consistency?
- How many *times* am I rolling this?

TEAMMATES

THE CASINO ALWAYS WINS...

TEAMMATES

- You are a Tester, you work with Developers

THE CASINO ALWAYS WINS...

TEAMMATES

- You are a Tester, you work with Developers
- Each Tester can diagnose one bug a day

THE CASINO ALWAYS WINS...

TEAMMATES

- You are a Tester, you work with Developers
- Each Tester can diagnose one bug a day
- Each Developer rolls one dice a day

THE CASINO ALWAYS WINS...

How often do you find a bug?

How often do you find a bug?

Developer:

Works

Works

Works

Bug

How often do you find a bug?

Developer 1:

Works

Works

Works

Bug

Developer 2:

Works

Works

Bug

Works

Developer 3:

Works

Bug

Works

Works

Developer 4:

Bug

Works

Works

Works

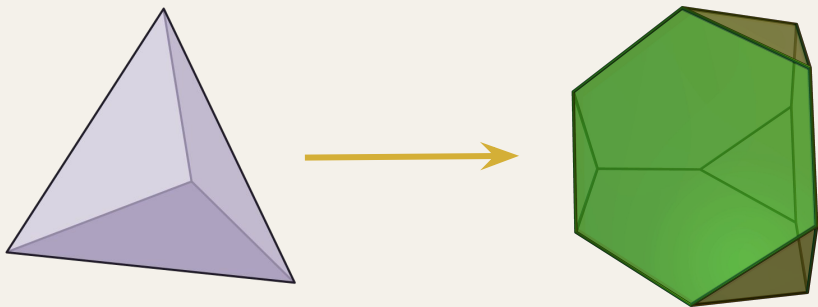
How can we reduce the bug rate?

How can we reduce the bug rate?

Cutting corners!

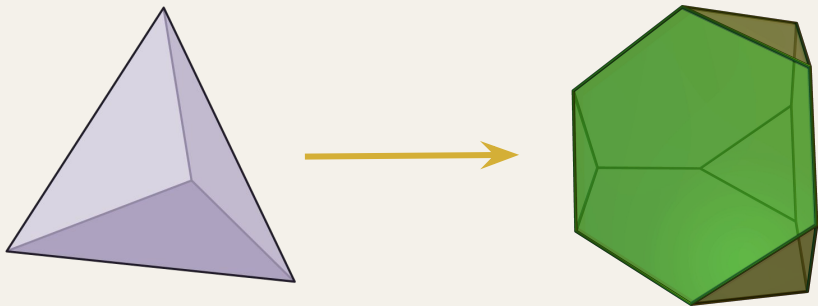
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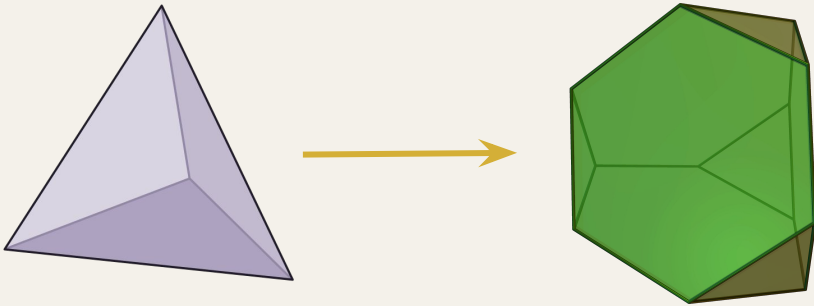
Cutting corners!



New Rules:

How can we reduce the bug rate?

Cutting corners!



New Rules:

“No parties ” → *“No merging
without
testing.”*

New Rule: No Bugs

New Rule: No Bugs

Works

Works

Works

Works

Works

Works

Works

Works

Works

Bug

New Rule: No Bugs

Works	Works	Works	Works	Works	Works	Works	Works	Works	Bug
Works	Works	Works	Bug	Works	Works	Works	Works	Works	Works
Works	Bug	Works	Works	Works	Works	Works	Works	Works	Works
Works	Works	Works	Bug	Works	Works	Works	Works	Works	Works

New Rule: No Bugs

Works	Works	Works	Works	Works	Works	Works	Works	Works	Bug
Works	Works	Works	Bug	Works	Works	Works	Works	Works	Works
Works	Bug	Works	Works	Works	Works	Works	Works	Works	Works
Works	Works	Works	Bug	Works	Works	Works	Works	Works	Works

(Tuesday)

TEAMMATES

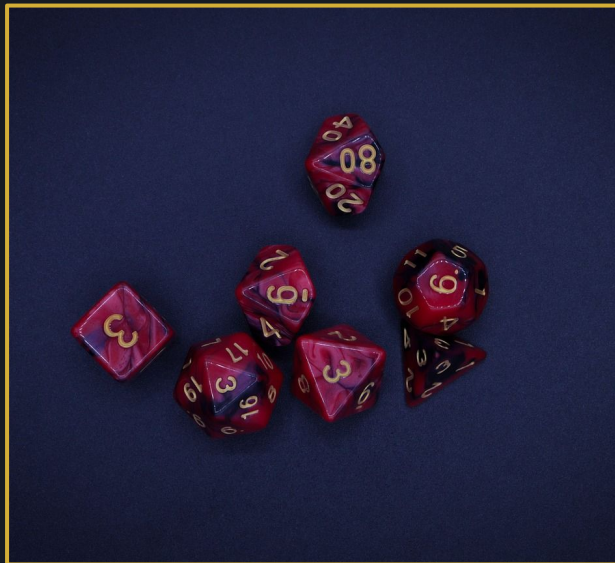
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TEAMMATES

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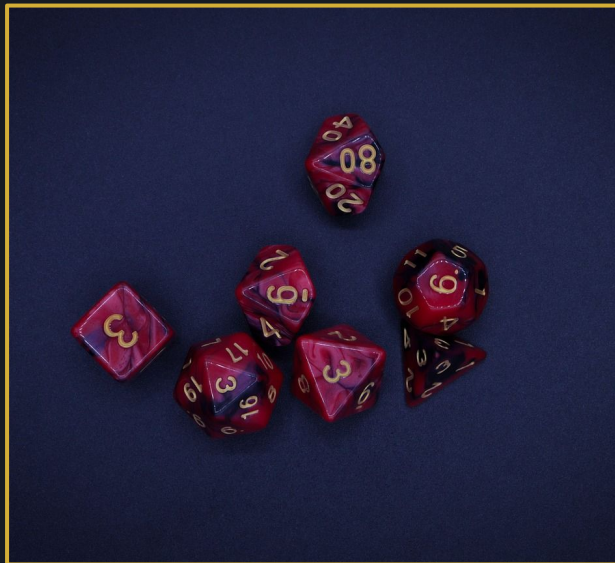
- Bugs are an *outcome* of development



TEAMMATES

How can we Lower the Bug Rate?

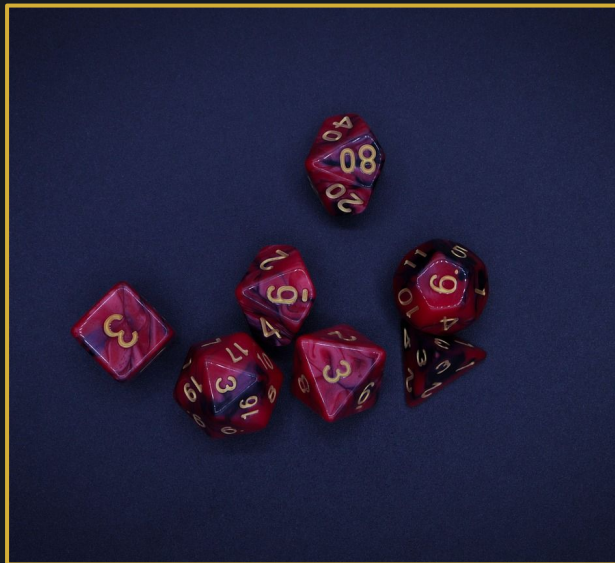
- Bugs are an *outcome* of development
- Breathing room is *essential* to balance



TEAMMATES

How can we Lower the Bug Rate?

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- The lower the *chance*, the better it scales



TEAMMATES

How can we Lower the Bug Rate?

- Bugs are an *outcome* of development
- Breathing room is *essential* to balance
- The lower the *chance*, the better it scales
- Testing is a way to *create* room



RELEASING

GOING ALL IN...

RELEASING

- There is a non-reproducible bug report

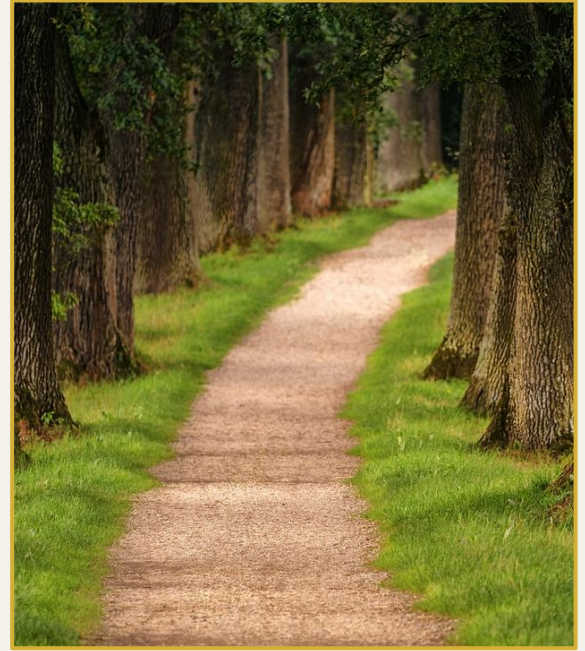
GOING ALL IN...

RELEASING

- There is a non-reproducible bug report
- Every behavior is a face on the dice

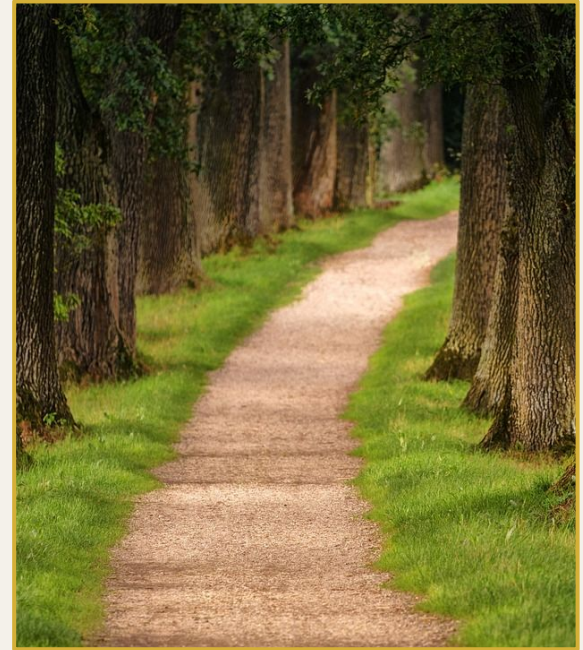
GOING ALL IN...

What can we learn about this dice?



What can we learn about this dice?

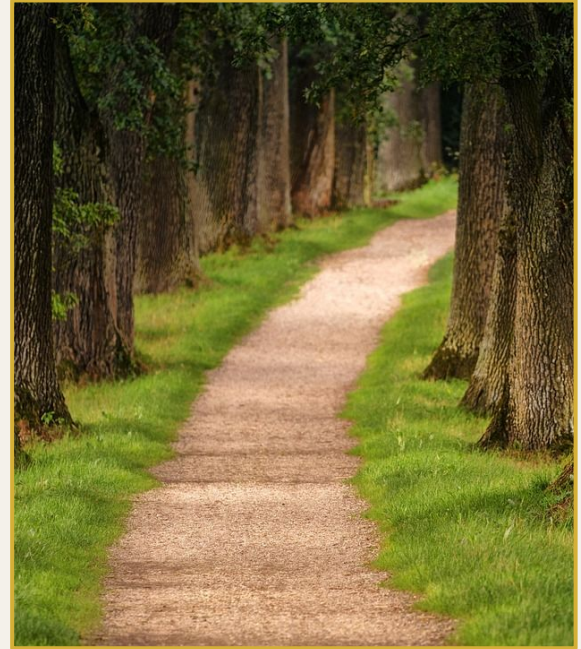
What are the known faces?



What can we learn about this dice?

What are the known faces?

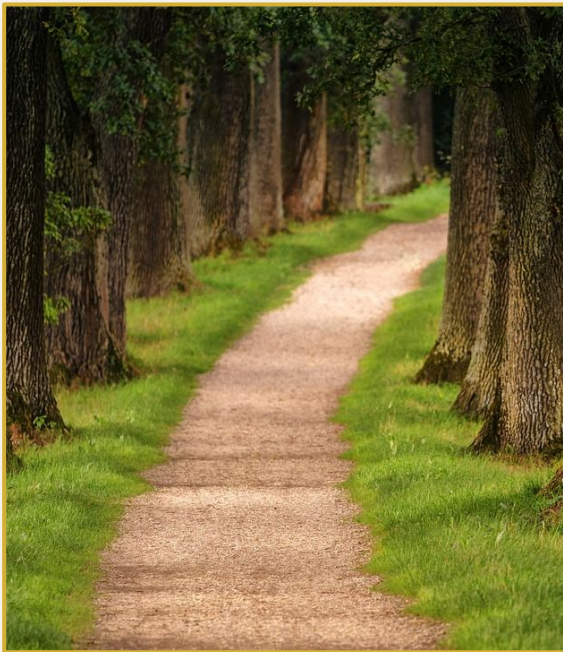
- What automated test coverage exists?



What can we learn about this dice?

What are the known faces?

- What automated test coverage exists?
- Are there any documented bugs?

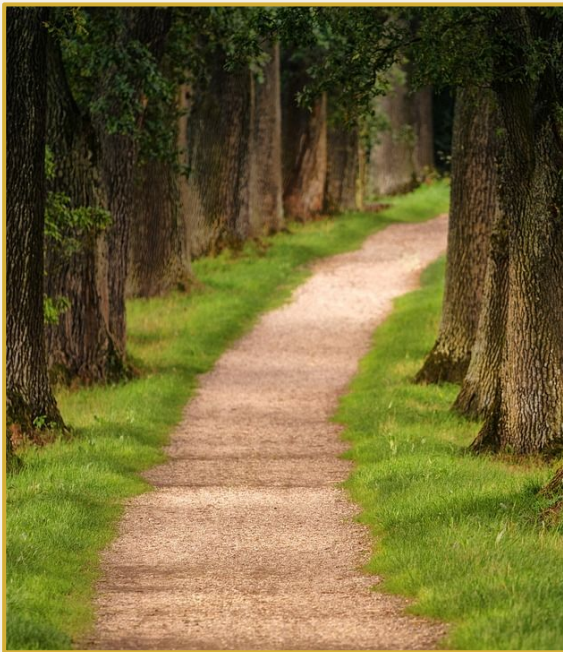


What can we learn about this dice?

What are the known faces?

- What automated test coverage exists?
- Are there any documented bugs?

What unknowns can we explore?



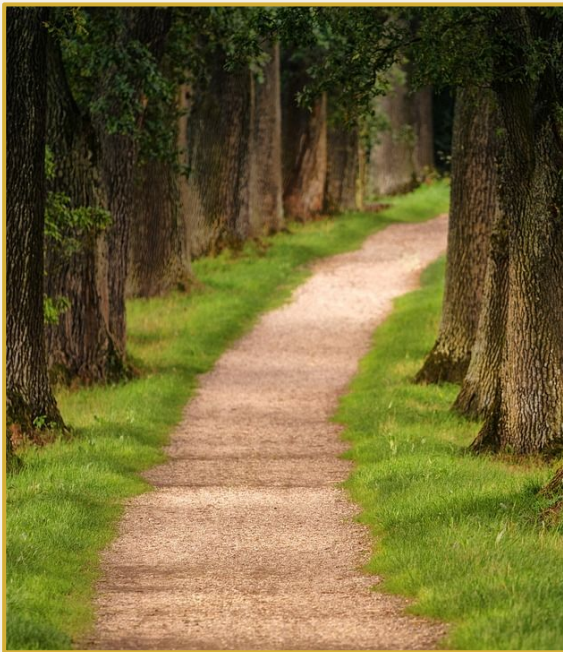
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What unknowns can we explore?

- Spot checks, manual testing



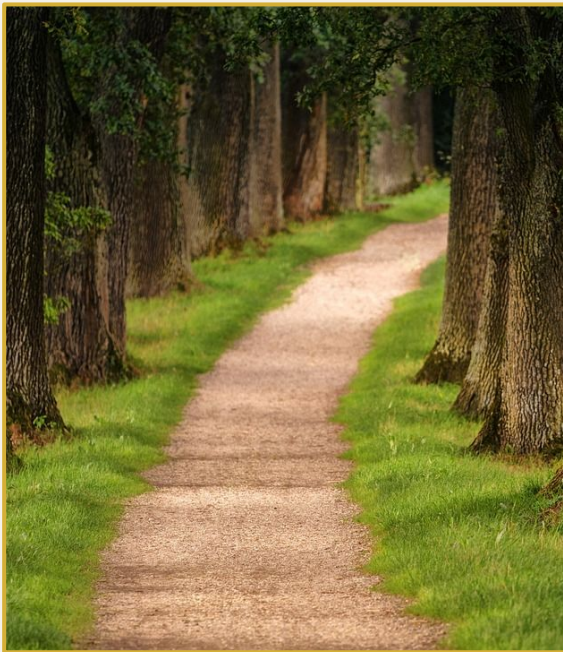
What can we learn about this dice?

What are the known faces?

- What automated test coverage exists?
- Are there any documented bugs?

What unknowns can we explore?

- Spot checks, manual testing
- End-to-end workflows



What can't we learn about this dice?



What can't we learn about this dice?



What unknowns can't we explore?

What can't we learn about this dice?



What unknowns can't we explore?

- Are there any flaky tests?

What can't we learn about this dice?



What unknowns can't we explore?

- Are there any flaky tests?
- What testing technology are we lacking?

What can't we learn about this dice?



What unknowns can't we explore?

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- What testing technology are we lacking?

What possibilities could be left?

What can't we learn about this dice?



What unknowns can't we explore?

- Are there any flaky tests?
- What testing technology are we lacking?

What possibilities could be left?

- What didn't we have time to test?

What can't we learn about this dice?



What unknowns can't we explore?

- Are there any flaky tests?
- What testing technology are we lacking?

What possibilities could be left?

- What didn't we have time to test?
- How creative have we been with testing?

What are we giving to customers?



What are we giving to customers?

*What risk is your
customer willing to
accept?*



What are we giving to customers?

What risk is your customer willing to accept?



What behaviors do your customer expect?

What are we giving to customers?

What risk is your customer willing to accept?

How is your thinking different from the customer's?



What behaviors do your customer expect?

What are we giving to customers?

What risk is your customer willing to accept?

How is your thinking different from the customer's?



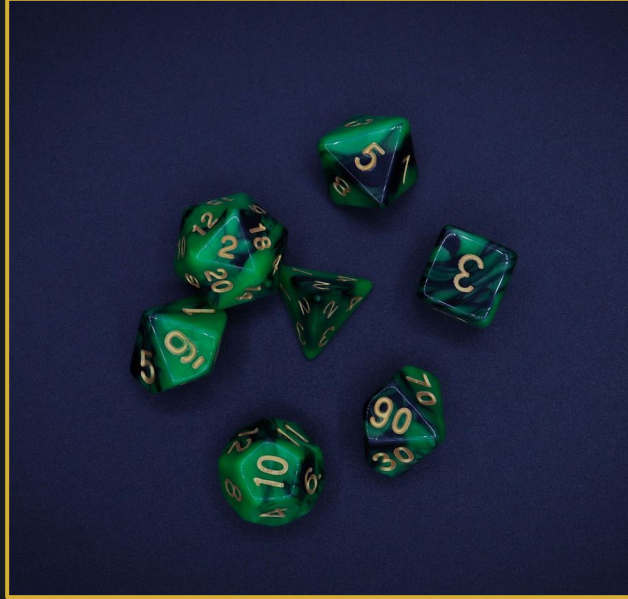
What behaviors do your customer expect?

Will your customer want to roll this?

RELEASES



How do we model a complex risk scenario with dice?



RELEASES

How do we model a complex risk scenario with dice?

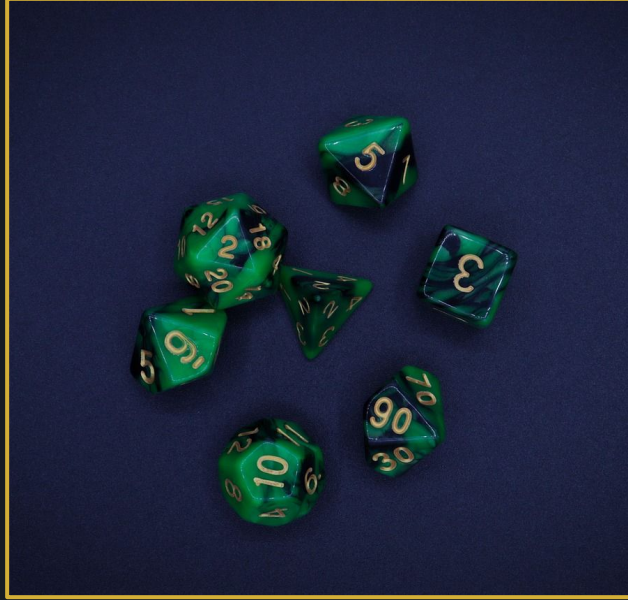
- We need to *uncover* what the dice looks like



RELEASES

How do we model a complex risk scenario with dice?

- We need to *uncover* what the dice looks like
- We need to *know* the unknown



RELEASES

How do we model a complex risk scenario with dice?

- We need to *uncover* what the dice looks like
- We need to *know* the unknown
- We're *delivering* the dice to the customer

DECIDING

WHEN DO YOU FOLD?

DECIDING

- Every day we roll many dice

WHEN DO YOU FOLD?

DECIDING

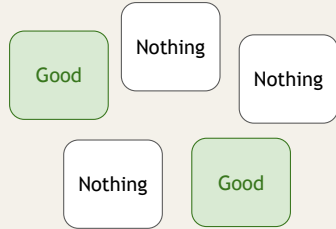
- Every day we roll many dice
- Do we like what these dice look like?

WHEN DO YOU FOLD?

What does your die look like?

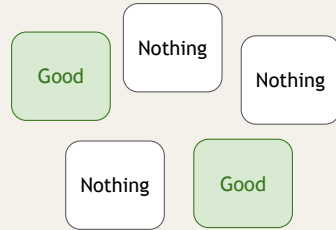
What does your die look like?

Mostly Positive?

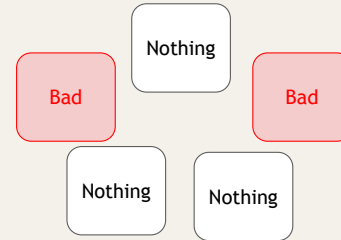


What does your die look like?

Mostly Positive?

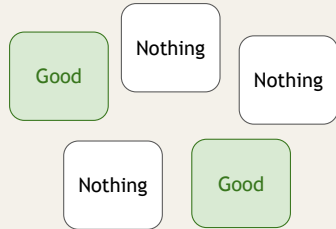


Mostly Negative?

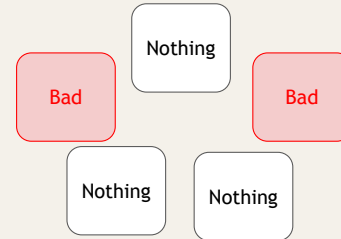


What does your die look like?

Mostly Positive?



Mostly Negative?



High Variance?

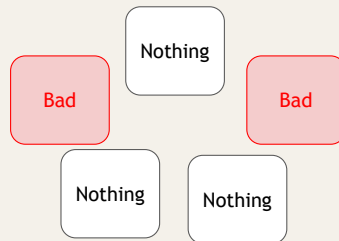


What does your die look like?

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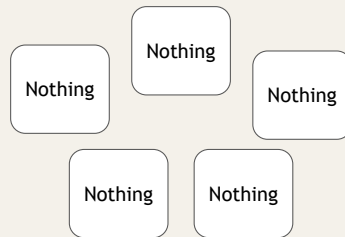
Mostly Negative?



High Variance?



Totally Neutral?

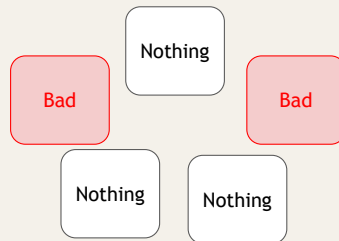


What does your career look like?

Mostly Positive?



Mostly Negative?



High Variance?



Totally Neutral?

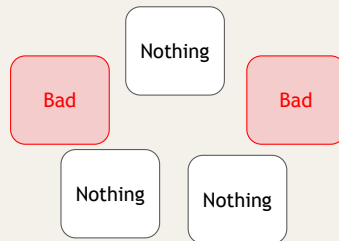


What does your career look like?

Mostly Positive?



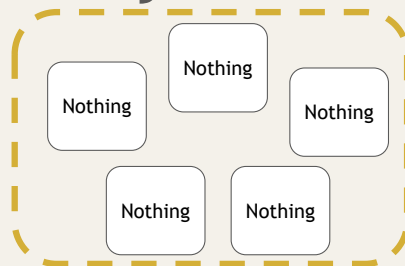
Mostly Negative?



High Variance?



Totally Neutral?



(Tuesday)

DECIDING

Are these dice
worth rolling?



DECIDING

Are these dice worth rolling?

- Decisions are about *comfort*, not numbers



DECIDING

Are these dice worth rolling?

- Decisions are about *comfort*, not numbers
- Everyone *perceives* faces differently





DECIDING

Are these dice worth rolling?

- Decisions are about *comfort*, not numbers
- Everyone *perceives* faces differently
- We might *eliminate* reward as we reduce risk



DECIDING

Are these dice worth rolling?

- Decisions are about *comfort*, not numbers
- Everyone *perceives* faces differently
- We might *eliminate* reward as we reduce risk
- There is always an *opportunity* cost

ROLLING THE DICE

ROLLING THE DICE

- Do the math - crunch the numbers

ROLLING THE DICE

- Do the math - crunch the numbers
- Chip away at functional areas with testing

ROLLING THE DICE

- Do the math - crunch the numbers
- Chip away at functional areas with testing
- Determine what you don't know

ROLLING THE DICE

- ▣ Do the math - crunch the numbers
- ▣ Chip away at functional areas with testing
- ▣ Determine what you don't know
- ▣ Consider your customer's risk tolerance

ROLLING THE DICE

- Do the math - crunch the numbers
- Chip away at functional areas with testing
- Determine what you don't know
- Consider your customer's risk tolerance



Rolling the Dice

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